

AI VISIBILITY AUDIT

AI Visibility Audit Report: https://zaavia.net/

Target Domain: https://zaavia.net/

Audit Date: August 20, 2026

Site Classification: B2B Software Development & Healthcare Solutions (Static S3 + Amazon CloudFront CDN)

Overall Readiness Verdict: **PARTIALLY READY**

Executive Summary

zaavia.net has **open, unrestricted crawler access** for all major AI search and answer-engine bots (GPTBot, ClaudeBot, PerplexityBot, Google-Extended, etc.), and its homepage copy is **server-rendered directly in HTML** rather than locked behind client-side JavaScript.

However, AI visibility is hindered by a **complete absence of structured data (JSON-LD)**, a **missing XML sitemap** (404), a **missing llms.txt** (404), and **missing origin security headers** on CloudFront.

Core Signals Scorecard

DIMENSION	STATUS	KEY FINDING	SPECIALIST DELEGATE
Crawler Access & Robots	PASS	All 8 major AI crawlers return HTTP 200 with full content.	robots-ai-crawler-audit
Sitemap & Discovery	FAIL	https://zaavia.net/sitemap.xml returns 404; missing in robots.txt.	sitemap-discovery-audit
Structured Data (JSON-LD)	FAIL	0 JSON-LD blocks found; missing Organization and Service schemas.	schema-markup-audit
On-Page Context & Headings	PASS	Clear H1 (Software Development Company Karachi- Pakistan) and 5 H2 sections.	answer-engine-content-audit
Citation & Entity Trust	PARTIALLY READY	Public claims present, but missing schema entity linkage.	citation-readiness-audit
Origin Security Headers	FAIL	Missing HSTS, X-Content-Type-Options, and X-Frame-Options.	robots-ai-crawler-audit
Image Discoverability	PASS	38 images sampled; 100% carry valid alt text attributes.	image-audit
Machine Context (llms.txt)	FAIL	https://zaavia.net/llms.txt returns 404 Not Found.	llms-txt-generator

Top Blockers

#	BLOCKER	SEVERITY	IMPACT SUMMARY	SPECIALIST DELEGATE
1	Zero Structured Data (JSON-LD)	CRITICAL	AI search engines cannot extract unambiguous machine-readable entity identity, services, or healthcare products.	schema-markup-audit
2	Missing XML Sitemap (/sitemap.xml 404)	HIGH	Search and agent crawlers cannot discover deep URLs or verify freshness.	sitemap-discovery-audit
3	Missing llms.txt (/llms.txt 404)	MEDIUM	Autonomous agents lack a curated index of authoritative product and docs URLs.	llms-txt-generator
4	Missing Origin Security Headers	MEDIUM	CloudFront lacks HSTS and MIME-sniffing protection headers.	robots-ai-crawler-audit
5	robots.txt Syntax Anomalies	LOW	Trailing spaces in Disallow: /users / etc. risk path matching bugs.	robots-ai-crawler-audit

Detailed Evidence & Observations

1. Crawler & User-Agent Access

- Observed Command: for ua in GPTBot ClaudeBot PerplexityBot Google-Extended CCBot Amazonbot; do ...; done
- Observed Output:

TEXT

```
GPTBot      Status: 200, Content-Length: 609380, Content-Type: text/html
OAI-SearchBot  Status: 200, Content-Length: 609380, Content-Type: text/html
ClaudeBot   Status: 200, Content-Length: 609380, Content-Type: text/html
PerplexityBot Status: 200, Content-Length: 609380, Content-Type: text/html
Google-Extended Status: 200, Content-Length: 609380, Content-Type: text/html
CCBot       Status: 200, Content-Length: 609380, Content-Type: text/html
Amazonbot   Status: 200, Content-Length: 609380, Content-Type: text/html
Googlebot   Status: 200, Content-Length: 609380, Content-Type: text/html
```

- Analysis: All major AI crawlers have full public access to the homepage HTML without WAF rate-limiting or captchas.

2. robots.txt & Sitemap Directives

- Observed Command: curl -s https://zaavia.net/robots.txt
- Observed Output:

```
User-agent: *
Disallow: /login/
Disallow: /users /
Disallow: /sessions /
Disallow: /logout /
Disallow: /signup /
```

- **Analysis:** Public marketing and content routes are open. However, trailing spaces exist on multiple `Disallow:` directives, and no `Sitemap:` reference is declared. Probing `https://zaavia.net/sitemap.xml` returns `HTTP 404` (`NoSuchKey` S3 error).

3. Machine-Readable Structured Data (Schema.org JSON-LD)

- **Observed Command:** `curl -s https://zaavia.net/ | grep -oE '<script type="application/ld\+json">[^\<]*'`
- **Observed Output:**

```
[No JSON-LD blocks found]
```

- **Analysis:** Zero Schema.org entities exist on the homepage. Knowledge graphs have no structured data to link Zaavia to its brand name, services, healthcare software solutions, or physical location.

4. On-Page Context, Meta Tags & Heading Hierarchy

- **Title Tag:** `Software Development Company Karachi- Pakistan` (48 chars)
- **Meta Description:** `Software Development Company in Pakistan with Global presence, specializing in Healthcare Software Development & Productivity Tools - Get FREE Demo TODAY!` (164 chars)
- **Canonical URL:** `https://zaavia.net/` (Self-referencing canonical verified)
- **Heading Structure:**
 - `<h1>`: `Software Development Company Karachi- Pakistan`
 - `<h2>`: `Trusted by Healthcare Professionals`, `Why Businesses Choose Zaavia`, `What We Do`, `Our Products`, `What Our Clients are Saying?`
- **Server Renderability:** 609 KB of content is served directly in the initial HTML payload, ensuring full visibility for non-JavaScript crawlers.

5. Origin Security Headers

- **Observed Command:** `curl -sI https://zaavia.net/`
- **Observed Output:**

```
HTTP/1.1 200 OK
Content-Type: text/html
Server: AmazonS3
X-Cache: Hit from cloudfront
Strict-Transport-Security: [MISSING]
X-Content-Type-Options: [MISSING]
X-Frame-Options: [MISSING]
```

- **Analysis:** CloudFront serves S3 objects without an attached Response Headers Policy for HSTS, MIME sniffing protection (`nosniff`), or clickjacking defense.

6. Image Alt Text & Discovery

- **Sample Scope:** 38 images sampled across homepage
- **Missing / Empty Alt Count:** 0
- **Analysis:** Excellent compliance. All product logos, client badges, and hero imagery contain non-empty descriptive `alt` attributes.

⚡ Quick Wins

1. **Fix** `robots.txt` **trailing spaces:** Clean up `Disallow: /users/` , `/sessions/` , `/logout/` , `/signup/` .
2. **Publish** `sitemap.xml` : Deploy an XML sitemap to the S3 bucket and declare `Sitemap:` `https://zaavia.net/sitemap.xml` in `robots.txt` .
3. **Add** `Organization` & `Service` **JSON-LD:** Embed Schema.org JSON-LD on the homepage describing Zaavia, its healthcare software, and services.
4. **Draft & Deploy** `/llms.txt` : Place an `llms.txt` file at the root summarizing Zaavia's core services, products, and contact pages.
5. **Enable CloudFront Response Headers Policy:** Attach AWS managed `SecurityHeadersPolicy` to enforce HSTS and `X-Content-Type-Options: nosniff` .

EXPERIMENTAL Emerging Agent Signals (Draft Standards)

i *Note: These checks evaluate emerging draft standards surfaced by isitagentready.com and agent discovery working groups. Absence of these signals does **not** harm search engine discovery, crawler indexing, or AI platform visibility today.*

- **Markdown Content Negotiation** (`Accept: text/markdown`): **FAIL** — Server returns `Content-Type: text/html` (standard HTML).
- **Content Signals in** `robots.txt` : `MISSING` — No `Content-Signal:` header or directive declared.
- **Web Bot Auth** (`/.well-known/bot-auth`): `404` — Cryptographic bot authentication not deployed.
- **Agential Resource Discovery** (`/auth.md` , `/.well-known/ard.json`): `404` — Draft manifest files not published.
- **DNS-AID** (`_aid.zaavia.net`): `MISSING` — No TXT discovery record declared.

Prioritized Remediation Roadmap

Ticket 1: Implement Schema.org JSON-LD Structured Data

- **Priority:** **CRITICAL**
- **Delegate Skill:** schema-markup-audit
- **Scope & Action:** Add `Organization` and `ProfessionalService` JSON-LD to `index.html` with `name`, `url`, `logo`, `description`, `sameAs`, and `contactPoint`.
- **Verification Command:**

```
curl -s https://zaavia.net/ | grep -q 'application/ld+json'
# expect: 0 exit code (was: not found)
```

Ticket 2: Generate and Publish XML Sitemap

- **Priority:** **HIGH**
- **Delegate Skill:** sitemap-discovery-audit
- **Scope & Action:** Upload valid `sitemap.xml` to S3 root and add `Sitemap: https://zaavia.net/sitemap.xml` to `robots.txt`.
- **Verification Command:**

```
curl -s -o /dev/null -w "%{http_code}\n" https://zaavia.net/sitemap.xml
# expect: 200 (was: 404)
```

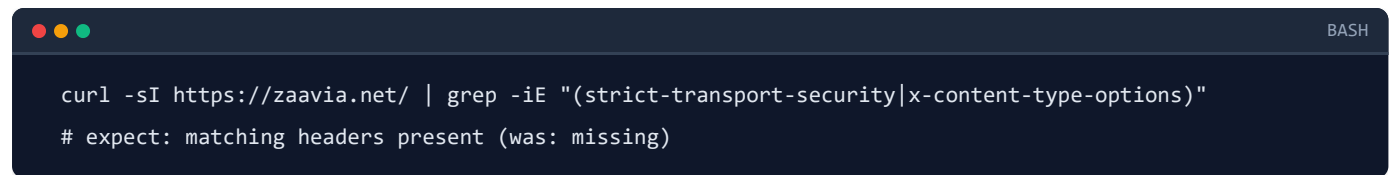
Ticket 3: Draft and Deploy `llms.txt`

- **Priority:** **MEDIUM**
- **Delegate Skill:** llms-txt-generator
- **Scope & Action:** Create `/llms.txt` linking to Zaavia's products, healthcare solutions, and contact channels.
- **Verification Command:**

```
curl -s -o /dev/null -w "%{http_code}\n" https://zaavia.net/llms.txt
# expect: 200 (was: 404)
```

Ticket 4: Configure CloudFront Security Headers & Fix `robots.txt`

- **Priority:** **MEDIUM**
- **Delegate Skill:** robots-ai-crawler-audit
- **Scope & Action:** Remove whitespace typos in `robots.txt` Disallow lines, and enable CloudFront `SecurityHeadersPolicy` (`Strict-Transport-Security`, `X-Content-Type-Options: nosniff`).
- **Verification Command:**



```
curl -sI https://zaavia.net/ | grep -iE "(strict-transport-security|x-content-type-options)"  
# expect: matching headers present (was: missing)
```